

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (CCE) (Credit System) Degree Examinations**

April 2023

21CC304 – DATA COMMUNICATIONS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I**Marks BT* CO* PO***

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|-------|---|---|-----|---|---|
| 1. a) | Define Data Communication. Explain simplified communication model of computer network with neat diagram. | 8 | L*2 | 1 | 1 |
| b) | Suppose that the spectrum of a channel is between 3MHz and 4 MHz and $SNR_{dB} = 24dB$. Find the following: | | | | |
| | i) SNR, | | | | |
| | ii) Capacity of the channel and | | | | |
| | iii) How many signaling levels are required? | 6 | L1 | 2 | 2 |
| c) | Given the dataword 101001111 and the divisor 10111, apply binary division and show the generation of the CRC codeword at the sender site. | 6 | L3 | 2 | 2 |
| 2. a) | What is computer network? List and explain three major types of computer networks based on the size of the network. | 8 | L2 | 1 | 1 |
| b) | Explain four categories of noises. In a communication channel, the power of a signal is 10 mW and the power of the noise is 1μW; what are the values of SNR and SNR_{dB} ? | 6 | L2 | 2 | 2 |
| c) | Explain error correction process with block diagram. A code scheme has a Hamming distance $d_{min} = 6$. | | | | |
| | i) What is error detection capability? | | | | |
| | ii) What is error correction capability? | 6 | L2 | 2 | 2 |
| 3. a) | Define Internet. Explain architecture of an internet by describing internet terminologies with suitable diagram. | 8 | L2 | 1 | 1 |
| b) | Explain digitization process based upon sampling theorem with a block diagram. | 6 | L2 | 2 | 2 |
| c) | Apply traditional internet checksum to find the checksum for the following set of data $(0001)_{16}$ $(F203)_{16}$ $(F4F5)_{16}$ $(F6F7)_{16}$. Show how receiver verifies the checksum. | 6 | L3 | 2 | 2 |

Unit – II

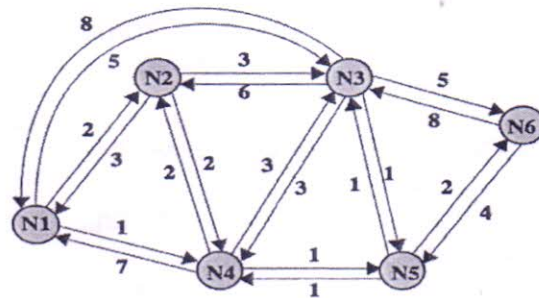
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|-------|--|---|----|---|---|
| 4. a) | With a neat diagrammatic depiction, explain the concept of sliding window in flow control. | 8 | L2 | 3 | 2 |
| b) | Briefly explain TDM. Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (i) the duration of 1 bit before multiplexing, (ii) the transmission rate of the link, (iii) the duration of a time slot, and (iv) the duration of a frame. | 8 | L2 | 3 | 2 |
| c) | Develop Dijkstra's algorithm for shortest path routing. | 4 | L3 | 4 | 2 |
| 5. a) | Briefly explain FDM. | 4 | L2 | 3 | 2 |
| b) | Explain time-slot interchange mechanism applied in time-division switches. | 8 | L2 | 3 | 2 |
| c) | List four routing strategies applied in packet switched networks. Explain ANY TWO in brief. | 8 | L2 | 4 | 2 |

P.T.O.

6. a) Summarize the differences between Go-Back-N and Selective Repeat ARQ protocol.
 b) List the advantages of packet switching networks over circuit switching. Explain two packet switching approaches used in contemporary networks.
 c) Apply Bellman-Ford algorithm to find the shortest path from source node 'N1' in the following network graph.

6 L2 3 2

8 L2 4 2



6 L3 4 2

Unit – III

7. a) Why synchronous access control is not suitable for LAN. Explain three types of asynchronous access control technique in MAC.
 b) List three Ethernet media access control protocols. Explain persistent methods of CSMA protocol.
8. a) Explain each fields of IEEE 802.3 MAC Frame with a neat frame format.
 b) With a neat IEEE 802.11 Protocol Architecture, explain DCF and PCF for medium access control.

10 L2 5 2

10 L2 5 2

10 L2 5 2

10 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
